

Algebra II with Trigonometry

Chapter 5.1

Olympiad Solutions

Saved tutoring response with MathJax rendering. Original PDF file:

[4401267_alg-2trig-h-chp-5-1-note-docx.pdf](#)

Solutions

1. Suppose a circle of radius r has a sector with central angle θ (in radians) whose area equals the arc length of the same sector. Find, in terms of r , all positive values of θ that satisfy this property.

Solution.

- **Sector area:** $\frac{1}{2}r^2\theta$
- **Arc length:** $r\theta$

Set equal:

$$\frac{1}{2}r^2\theta = r\theta$$

If $\theta > 0$, divide both sides by $r\theta$:

$$\frac{1}{2}r = 1 \implies r = 2$$

But we're asked for values of θ in terms of r . Alternatively, suppose the area equals the arc length:

$$\frac{1}{2}r^2\theta = r\theta$$

If $r \neq 0$ and $\theta \neq 0$,

$$\frac{1}{2}r\theta = \theta \implies r = 2$$

Thus, for any positive θ , $r = 2$ is required, not the other way.

But perhaps the intended meaning is that, **for given r , find all positive θ** that make the equality hold. So,

$$\frac{1}{2}r^2\theta = r\theta \implies \frac{1}{2}r^2\theta - r\theta = 0 \implies r\theta \left(\frac{1}{2}r - 1 \right) = 0$$

Since $r, \theta > 0$, this means $r = 2$.

Thus, for all positive values of θ , the equality can only happen if $r = 2$.

Or perhaps you want θ on both sides: area = arc length:

$$\frac{1}{2}r^2\theta = r\theta'$$

But it asks for **values of θ** , so that's clear: **no positive value of $\theta \neq 0$ works unless $r = 2$** ; if $r = 2$, any positive θ works.

Answer:

All positive values of θ , but only if $r = 2$.

If r is arbitrary and fixed, there are **no positive θ** unless $r = 2$, otherwise none.

2. How many distinct positive integers n are there such that n° and $2n^\circ$ are coterminal angles in standard position?

Solution.

n° and $2n^\circ$ are coterminal if their difference is a multiple of 360° :

$$2n - n = n \equiv 0 \pmod{360} \implies 360 \mid n$$

So positive integers $n \leq 360$ divide 360.

Number of positive divisors of 360:

$360 = 2^3 \cdot 3^2 \cdot 5^1$, so number of divisors:

$$(3 + 1)(2 + 1)(1 + 1) = 4 \times 3 \times 2 = 24$$

Answer:

24

3. The minute and hour hand of a clock are both 10 cm long. At exactly 12:00, both hands are superimposed. After one hour, what is the exact total area traced out by the tips of the hands, treating each path as a sector swept on a circle of radius 10 cm? Express your answer in terms of π .

Solution.

- **Minute hand:** Completes one full revolution (2π radians) in one hour.
- **Hour hand:** Moves from 12:00 to 1:00. In 1 hour, it moves $\frac{1}{12}$ of a circle, i.e., $\frac{2\pi}{12} = \frac{\pi}{6}$ radians.
- **Area traced by the tip = area of sector:** $\frac{1}{2}r^2\theta$, with $r = 10$ cm.

For the minute hand (full circle):

$$A_m = \frac{1}{2}(10^2)(2\pi) = 50 \times 2\pi = 100\pi \text{ cm}^2$$

For the hour hand:

$$A_h = \frac{1}{2} \times 10^2 \times \frac{\pi}{6} = 50 \times \frac{\pi}{6} = \frac{25\pi}{3} \text{ cm}^2$$

- **Total area:**

$$100\pi + \frac{25\pi}{3} = \frac{300\pi + 25\pi}{3} = \frac{325\pi}{3}$$

Answer:

$$\boxed{\frac{325\pi}{3} \text{ cm}^2}$$

4. Let A and B be two angles (in degrees), both less than 360° , such that the arc lengths subtended by A in a circle of radius r and by B in a circle of radius $2r$ are equal. If $A : B = 3 : 1$, what is the value of A ?

Solution.

- Arc length for A on radius r : $\ell_A = r \cdot \left(\frac{\pi}{180} A\right)$
- Arc length for B on radius $2r$: $\ell_B = 2r \cdot \left(\frac{\pi}{180} B\right)$
- Equate: $r \cdot \frac{\pi}{180} A = 2r \cdot \frac{\pi}{180} B \implies A = 2B$

But, $A : B = 3 : 1 \implies A = 3k, B = k$. But above,
 $A = 2B \implies 3k = 2k \implies k = 0$.

This suggests a contradiction. Let's check the equation:

Alternatively, perhaps we should check the equality:

$$r \left(\frac{\pi}{180} A \right) = 2r \left(\frac{\pi}{180} B \right) \implies A = 2B.$$

But given $A : B = 3 : 1$, so $A = 3x, B = x$, thus $3x = 2x \implies x = 0$.
 Contradicts $x > 0$. Alternatively, perhaps the correct ratio is "the length subtended by A in radius r " to "the length subtended by B in radius $2r$ " is **1:1**, but with ratio **3 : 1** for the angles.

Is there a misinterpretation? Or perhaps the question expects another approach.

Suppose $A = 3B$:

Then,

$$r \cdot \frac{\pi}{180} A = 2r \cdot \frac{\pi}{180} B \implies A = 2B$$

But also $A = 3B$. So the only solution is $B = 0$.

But both angles must be less than 360° , and positive.

Therefore, perhaps no such value.

Alternatively, perhaps the intended meaning is "if the ratio $A : B = 3 : 1$ ", what is A such that arc lengths are equal.

That is:

$$A = 3x, B = x$$

Arc length equality:

$$r \cdot \frac{\pi}{180}(3x) = 2r \cdot \frac{\pi}{180}(x) \implies 3x = 2x \implies x = 0$$

No positive solution.

Conclusion: The only values are zero, which are excluded. Possibly a typo in the problem.

However, perhaps the question was meant for ratios $1 : 3$ or $2 : 1$. In the current wording, **no positive solution exists**.

5. A Ferris wheel with radius 15 m completes one full counterclockwise revolution every 40 seconds. Another identical Ferris wheel next to it rotates in the opposite direction, making exactly n revolutions during the same amount of time. If the points at the topmost positions of the wheels align every 10 seconds, what is the value of n ?

Solution.

Let their angular speeds be $\omega_1 = \frac{2\pi}{40} = \frac{\pi}{20}$ (counterclockwise),
 $\omega_2 = -n \frac{2\pi}{40} = -\frac{n\pi}{20}$ (clockwise, negative).

Topmost point alignment means their angular displacement **sum** equals an integer multiple of 2π :

Let t be the time elapsed (in seconds), at which their top points align:

$$\omega_1 t + |\omega_2| t = 2\pi k, \quad \text{for some integer } k$$

Here, $|\omega_2| = \frac{n\pi}{20}$.

$$\left(\frac{\pi}{20} + \frac{n\pi}{20}\right)t = 2\pi k \implies \frac{(1+n)\pi}{20}t = 2\pi k \implies (1+n)t = 40k$$

Given that this happens every $t = 10$ seconds, so the smallest such t :

Set $t = 10$,

$$(1+n) \cdot 10 = 40k \implies (1+n) = 4k$$

Both n and k are positive integers, and in 40 seconds, the second wheel must complete exactly n revolutions (n integer, $t = 40$ is a period).

But with $t = 10$, $(1+n) = 4k \rightarrow n = 4k - 1$, for integer $k \geq 1$.

Now, $t = 40$ seconds must also be a time at which alignment occurs, i.e. 40 must be a multiple of 10.

Let's check what value of n makes n integer when $t = 40$:

$$n = 4k - 1$$

Set $t = 40$: this must align; what integer k yields that the second wheel's number of revolutions is an integer in 40 seconds? Use the other constraint:

In 40 seconds, second wheel rotates n times, so n is integer.

Let us check $k = 1, 2, 3, 4$:

- $k = 1: n = 3$
- $k = 2: n = 7$
- $k = 3: n = 11$
- $k = 4: n = 15$

Since n must also result in the wheels aligning every 10 seconds, and the period is 40 seconds:

Let's check if $n = 3$ satisfies this:

For $n = 3$:

- In 40 s, second wheel makes 3 revolutions (as required).

- Every 10 s, $t = 10$, $(1 + 3) \cdot 10 = 40$, so $k = 1$, so yes, every 10 s they align.

Thus,

Answer:

3

6. The area of a sector of a circle is numerically equal (in the same units) to five times the circumference of the circle. Find the central angle of the sector in radians, in terms of r .

Solution.

- **Circumference:** $2\pi r$
- **Sector area:** $\frac{1}{2}r^2\theta$

Given:

$$\frac{1}{2}r^2\theta = 5 \cdot 2\pi r = 10\pi r$$

Divide both sides by r ($r > 0$):

$$\frac{1}{2}r\theta = 10\pi \implies r\theta = 20\pi \implies \theta = \frac{20\pi}{r}$$

Answer:

$$\theta = \frac{20\pi}{r}$$